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A Simulation-Driven Framework for 3D Completion Quality Scoring in Hydraulic Fracturing

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Abstract

This paper introduces a simulation-driven framework aimed at improving large-scale reservoir screening in hydraulic fracturing. The method integrates static geological and geomechanical attributes (e.g., stress, rock strength, structural features, natural fracture density), along with dynamic fracture behavior derived from ultra-fast simulations at the field scale. Properties such as the containment, conductivity, and geometry of the induced fractures are aggregated into a unified, spatially resolved 3D completion quality cube—a volumetric representation of the reservoir that encodes a composite score at each location. The approach is illustrated through application to a real reservoir scenario, demonstrating its ability to identify favorable regions for stimulation based on both static and dynamic subsurface characteristics.

Introduction

The success of hydraulic fracturing is governed by a complex interplay of variables acting across multiple spatial and temporal scales. During operations, engineers must make critical decisions regarding fluid composition, proppant type, and pumping schedule, among others. A range of numerical simulators exists to model fracture growth in detail on a stage-by-stage basis, offering valuable insights for well-specific design. However, these well-centric approaches are most effective only after well locations have been established. At earlier stages of field development, the decision of where to drill is often informed by prior assumptions about large-scale geological and geomechanical features. Variables such as the in-situ stress field, structural heterogeneities, and the distribution and state of natural fractures are commonly used as proxies for stimulation potential. For instance, the density and criticality of natural fractures are frequently associated with the expected stimulated reservoir volume or fracture complexity [1, 2]. Nevertheless, regions that appear favourable based on natural fracture characteristics may be less attractive when considering local stress conditions (i.e. anisotropy, minimum stress magnitude). Hydraulic fracture containment, for example, is influenced by stress contrasts between subsurface layers (stress barriers). Yet, the relationships between stress barriers, fracture containment, conductivity and fracture propagation are not always straightforward.

Other variables such as the anisotropy and magnitude of the stress field can correlate with the complexity of the fractures.

Altogether, no single variable determines the ultimate success of fracturing or the best potential location for wells. Previous work [3, 4, 5] proposed a weighted aggregation of several geomechanical variables—used as proxies for the expected quality of hydraulic fracturing—into a single completion score. Yet, when relying solely on static properties of the subsurface, it is not trivial to assess the relative weight of each contribution, and some ad-hoc choices are inevitably required. This is the case, for example, when estimating the energy required to propagate a fracture through heterogeneous rock, evaluating the influence of stress barriers, or comparing the potential fracture conductivity across different reservoirs and locations. These are dynamic aspects of fracture propagation that can be better estimated by modeling the fracturing process itself.

In this work, we propose an extension of an existing workflow presented in [3, 4, 5] by incorporating a dynamic proxy for hydraulic fracture behaviour into the generation of a 3D *completion quality cube*. Our approach leverages ultra-fast hydraulic fracture simulations at potentially hundreds of thousands of reservoir locations to compute dynamic metrics such as fracture containment and conductivity directly from first principles. These outputs, combined with static properties derived from a high-quality mechanical earth model (e.g., fracability, tortuosity, natural fracture density) and any other user-defined attributes, are aggregated using user-defined weights into a single 3D quality score. The proposed technology complements traditional well-centric workflows by enabling early-stage, field-wide evaluation of completion quality, even before well placement decisions are made. A key component of this method is the generation of 3D quality maps that integrate both static and dynamic indicators, reducing the reliance on arbitrary proxies and offering a more physics-informed basis for optimizing pad placement and reservoir development.

The paper is organized in three main sections. First, the completion quality schema will be described. The following section shows an example of the application of the method. The last section summarizes our observations and conclusions.

Completion quality scheme

The Completion Quality (CQ) Scheme is a framework designed to evaluate, at the reservoir scale, the relative favorability of different locations for hydraulic fracturing while integrating multiple geomechanical and hydraulic indicators into a single spatial metric—the Completion Quality Cube (CQ Cube)—which serves as a composite score over the reservoir model volume.

Variable selection and normalization

The procedure starts by identifying a set of N input variables x_i representing a subsurface attribute expected to influence fracture initiation and propagation. Typical examples include breakdown pressure, stress anisotropy, natural fracture density and criticality, simulated fracture conductivity and containment.

Each variable x_i is mapped into a normalized variable $\hat{x}_i \in [0, 1]$ where the values of x_i that are more/less favorable map into the lower/upper range of $\hat{x}_i$. The result is a normalized cube $C_i(x, y, z)$ representing spatial favorability for variable x_i . As an example, one variable x could be the estimated conductivity of the hydraulic fracture. Assuming the higher the better, $\hat{x}$ would take values near one in the areas where the simulated fracture is more conductive and values near zero at the locations where the fracture conductivity is within the smaller values observed in the model (or reservoir). The mapping $x_i \rightarrow \hat{x}_i$ can be an arbitrary function (not necessarily linear). The process of constructing cubes for two example variables is sketched in Figure 1.

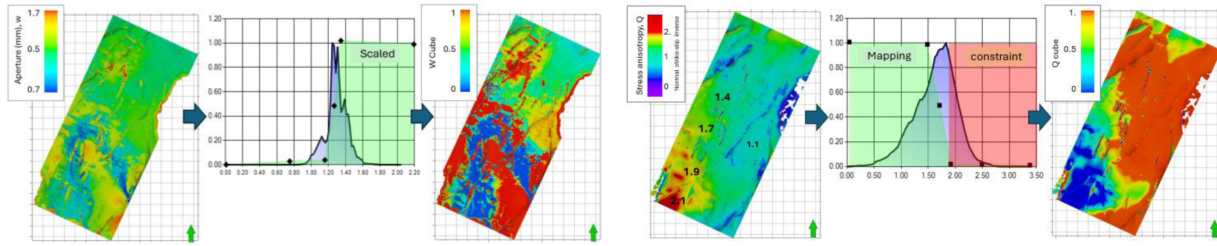


Figure 1—Example of completion cubes for two variables. In each case, the left colormap corresponds to the raw data. The distribution of values is shown in the histogram in the middle. The normalized cube in each case is shown at the right. The area curve -weighing- is the transformation applied (see text)

Figure 1 shows two example completion cubes: one for fracture conductivity (left), which can be computed via simulation of hydraulic fracturing (as we will explain later). The cube at the right refers to the stress anisotropy factor Q . Q is computed from the stress shape ratio $R = \frac{\sigma_2 - \sigma_3}{\sigma_1 - \sigma_3}$, where $\sigma_1 > \sigma_2 > \sigma_3$ are the principal stress components. The stress factor $Q \in [0, 3]$ is defined as $Q = R$, if the stress regime is normal (σ_1 is vertical), $Q = 2 - R$, for strike-slip regimes (vertical stress is the intermediate stress) and $Q = R + 2$ for inverse regimes (σ_3 is vertical). The principal stress components σ_i can be obtained from geomechanical modelling.

For the example case in Figure 1A and Figure 1B, the colormaps represent the raw values obtained from modelling/simulation. The distribution of values is shown in the histogram in the middle of each frame, the normalized cubes are shown at the right and the mapping function is shown with an area-filled curve overlying the histogram. In Figure 1A, such mapping function assigned values of zero to regions with apertures less than ≈ 1.2 mm and regions with apertures greater than the mean ($w \approx 1.3$ mm) were all given values of 1 in the normalized cube. A similar process is illustrated in the figure for Q . There, values of $Q > 2 \rightarrow 0$, $Q < 1 \rightarrow 1$ and for values in the range $[1.5, 2]$ Q decreases linearly from 1 to zero. In the figure it is shown that once such transformation is applied to the raw values of Q , a large area of the section is assigned the value of 1, as $Q < 1.5$ is considered favorable there according to the criterion represented in the mapping function. The Figure 1B also shows an extra curve labelled as "constraint". Constraints are imposed during the aggregation step discussed next.

Aggregation

Once the set of N normalized cubes $\{C_i\}$ are computed, we combine them into a single Completion Quality score at each location using a weighted and constrained summation:

$$s = \frac{1}{N} \Pi(f_i(x_i), g(y_i)) \sum w_i \hat{x}_i \quad (1)$$

$$s = \frac{1}{N} \Pi(f_i(x_i), g(y_i)) \sum w_i \hat{x}_i \quad (2)$$

where w_i is a discretionary weight in the range $[0, 1]$ given to each variable. The term $\Pi(\cdot)$ is the Gamma function (productory) and its arguments correspond to arbitrary constraints imposed over the x_i variables in the summation, or over arbitrary additional ones, y_i . For instance, y could be the stress regime (as an integer 1-normal, 2-strike-slip, 3-inverse) and $f(y)$ could be defined as a function equal to one in the range $[1, 2]$ and zero elsewhere. That way, one would assign a score of zero to any volume for which the stress regime is inverse. As defined, once any constrain function evaluates to zero at a given location, the aggregated score takes the value of zero at that location regardless of the potential contributions of other variables. Following with the example in Figure 1, a constraint is defined for the stress regime, i.e. $Q > 2 \rightarrow s = 0$. The aggregated score cube in this example is shown in Figure 2. Note that the final cube is always normalized in the range $[0, 1]$ and accounts for all the variables included in the analysis (only two in this example) and the constraints.

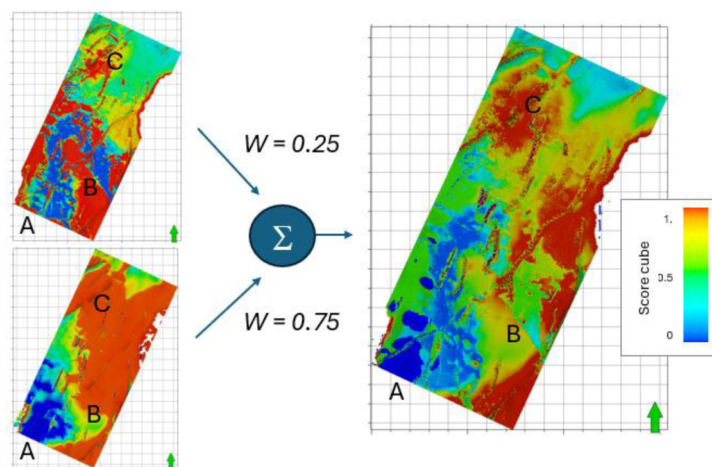


Figure 2—Aggregation first the example cubes in Figure 1

Note that in area A, the final score is zero due to the imposed constraint in Q . Around area B, the final score is around ≈ 0.5 as the criterion imposed for the aperture would lead to a maximum contribution, but the anisotropy is still high. The score cube takes values near 1 when both contributions are maximum.

In this example, we considered two variables for illustrative purposes. Yet any relevant subsurface attribute can in principle be accounted for if the data is available as a 3D cube. As an example, one could compute a cube representing the energy needed to fracture the rock by taking the 3D breakdown pressure cube as a proxy. It is possible to also generate other cubes after analyzing profiles of stiffness and σ_3 to represent the potential aperture of the fractures and their containment. Yet it isn't trivial to map those static variables to the ultimate characteristics of the fractures in the heterogeneous rock. It would be better to model the fracture propagation itself while accounting directly for the relevant variables. One challenge is, of course, producing 3D cubes of those attributes within a reasonable time so they can be incorporated in the completion quality schema.

Hydraulic fracturing proxy. In the early stages of field development planning, coarse yet physically grounded estimates of hydraulic fracturing behaviour—such as fracture geometry, aperture, and containment—can be sufficient to guide well placement decisions. At this stage, the objective is not to resolve fracture growth at the millimetre scale, as in high-resolution well-centric models, but rather to obtain reliable, first-order proxies that capture the essential behaviour of the system over large spatial domains.

To support this goal, we employ a semi-analytical hydraulic fracturing simulator derived from first principles, as detailed in [6, 7]. The model is built upon the framework of Lagrangian mechanics, where explicit expressions for potential energy, dissipation, and virtual work are used to derive equations governing key fracture geometry parameters, such as wing length and maximum aperture. Several simplifying assumptions reduce computational complexity while preserving the physical integrity of the system. For example, the fracture is assumed to adopt a rectangular shape with a constant aperture across its vertical profile. This aperture is not arbitrarily chosen but is estimated from planar elasticity, informed directly by the elastic properties of the surrounding rock and dynamic loading conditions. Proppant transport is not explicitly modelled; as such, the final fracture aperture represents an optimistic upper bound on the post-treatment propped width. Altogether, the set of equations and assumptions in the model is as follows (more details in [6]):

Fluid injection is assumed to be at a constant rate Q and the fluid is assumed to be a Newtonian fluid of viscosity μ . Material transport is described by the lubrication equation:

$$\frac{\partial p}{\partial x} = -12 \mu \frac{q_x}{w^3}, \quad \frac{\partial p}{\partial z} = -12 \mu \frac{q_z}{w^3}$$

Where p , q_z , q_x are the pressure, and flow rates along the vertical axis (z) and parallel axis to the fracture (x). The Carter's leakoff model described the fluid leakoff rate per unit fracture surface area:

$$u_L(x, t) = \frac{2C_L}{(t - \tau(x))^{1/2}}$$

Where C_L is the leakoff coefficient and $\tau \approx x/L$. The relation between pressure averaged over a fracture of average width follows:

$$\bar{p}(t) = \frac{\bar{E}}{2\pi} \bar{w} \frac{(L^2 + H^2)^{1/2}}{LH}$$

Where $H = H_t + H_b$, L are the height and length and H_t , H_b are respectively the top/bottom fracture length above/below the injection point. $\bar{p}$ is the net pressure average over H . The average plane strain modulus is given by $\bar{E} = (E'_t H_t + E'_b H_b) / H$, and $E'_i = E_i / (1 - \nu_i^2)$, where the subscript i , refers to either top or bottom and E_i , ν_i refers to the spatially averaged Young's modulus and Poisson's ratio. Further details on the solution of these equations can be found in [6] [7].

Altogether, these assumptions result in substantial computational speed, thus enabling rapid simulations across many locations while still enforcing energy conservation, mass balance, and asymmetric fracture propagation. This makes the approach ideally suited for high-throughput analyses, such as fracture screening over large fields, optimization of well spacing, real-time operational diagnostics, and pressure history matching. By focusing on scalable, physics-informed proxies rather than detailed geometrical predictions, the model provides a practical yet rigorous tool for early-phase decision making.

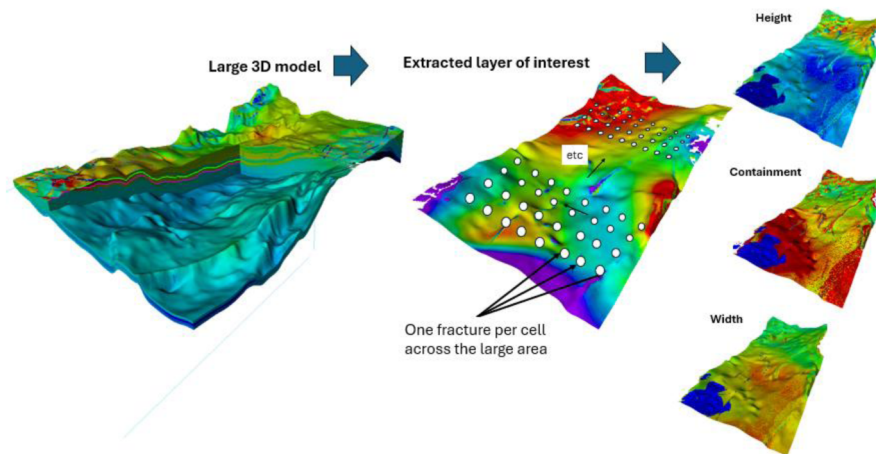


Figure 3—Large-scale modelling of hydraulic fractures. The figure at the left shows a 3D geomechanical model. The figure at the center shows a layer extracted from the model. There, a fracture is modelled in every cell. Results related to fracture propagation are then obtained for each location.

Application of the method

Field studied. The area under study is shale field in Colombia. There are several reservoirs stacked vertically. The results presented here correspond to the deepest reservoir, between the seismic horizons Pujama and Salada (see Figure 4). The reservoir is approximately 90ft to 130ft thick across the area under study. The lateral extent of the model is of the order of 20 km × 40 km. Clearly this is a large area and in principle, all of it is a candidate for hydraulic fracturing. The question we aim to answer is where the subsurface conditions are more favorable to place the first wells. This will be addressed using the

completion quality scheme defined before and while combining relevant geomechanical attributes and hydraulic fracturing attributes computed directly by simulation.

Geomechanical model. A mechanical Earth Model was built for this reservoir. There are main components in the model: The mechanical property model, the structural model and the stress model. The mechanical property model was built from well data guided by seismic using conventional methods for interpolation from well data (kriging guided by seismic). The size of the cells in the model grid were approximately 90 ft $\times$ 90 ft in the horizontal plane while it was refined along the vertical to have cells of the order of 2.5 ft thick. The geomechanical grid was oriented along the regional minimum stress, which was estimated to point 20 degrees North. The stress field itself was obtained in the 3D volume by finite element modelling. The simulator used to model stress has been used in several previous references, see for instance [8, 9]

A characteristic noted during modelling was an internal structure of the reservoir. Towards the South, the stiffness profiles at some wells showed relatively stiff layers, which in turn resulted in high stress concentration when modelling stress. Such an internal structure is present across the model, but towards the North, the rock appeared more homogeneous, which resulted in a nearly flat stress profile (see Figure 4). These high-stress layers can potentially act as barriers for fracture propagation, which will be investigated later in this paper.

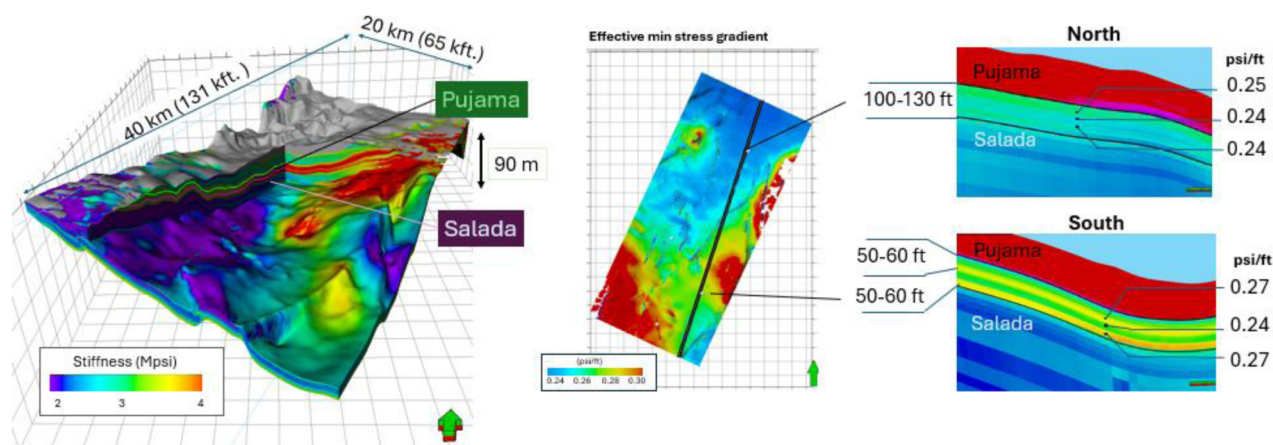


Figure 4—Snapshot of the model. Left: Model geometry Right: Detail of internal structures

Geomechanical attributes. The stress anisotropy was quantified via the stress factor Q (defined before). Figure 5 shows the computed value of Q and its distribution across a 2D layer vertically centered in the reservoir, located approximately in-between the two stiff layers shown in Figure 4. The breakdown pressure was computed using standard geomechanics formulae while assuming wells aligned with the regional direction of the regional minimum horizontal stress. The higher the breakdown pressure, the more energy is needed to initiate the fractures. Eventually, some pressures may be unattainable. Figure 5 shows 2D projections of the 3D results obtained.

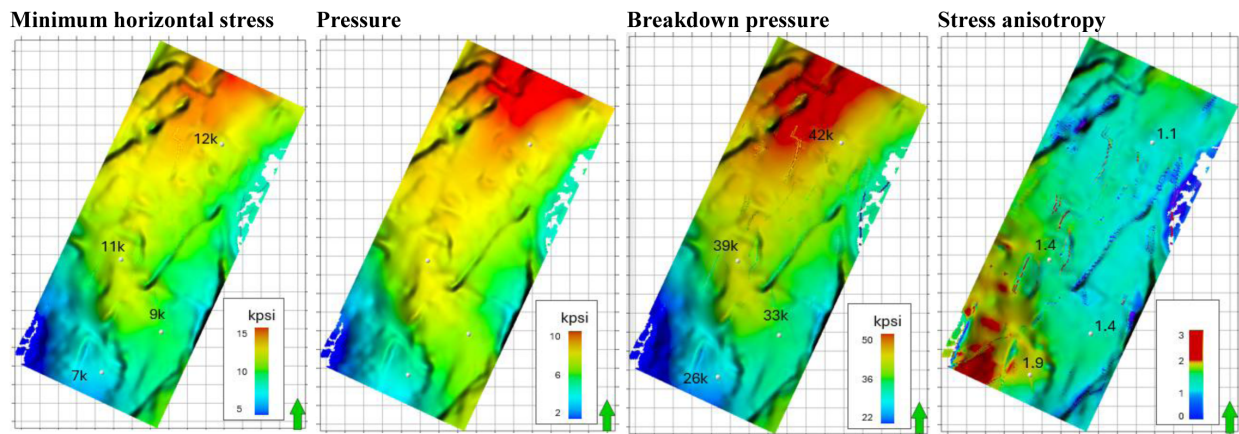


Figure 5—Geomechanical attributes across a later vertically centered in the reservoir.

Hydraulic fracturing attributes. The parameters used in the modelling of hydraulic fractures are shown in Table 1. The same parameters were used in all the simulations presented later. Results for fracture geometry are shown in Figure 6 and Figure 7.

Table 1—Stimulation parameters for the hydraulic fracturing simulation.

Parameter	Value	Units
Fluid density	1001/(1.001)	kg/m3(gr/cm3)
Fluid viscosity	1.0x10 ⁻³ /(1)	Pa /cP)
Pumping rate	0.1 /(37.7)	m3/s (bbl/min)
Pump volume	250 /(1500)	m3 (bbl)
Leak-ff coefficient	0.00035	m/s ^{0.5}

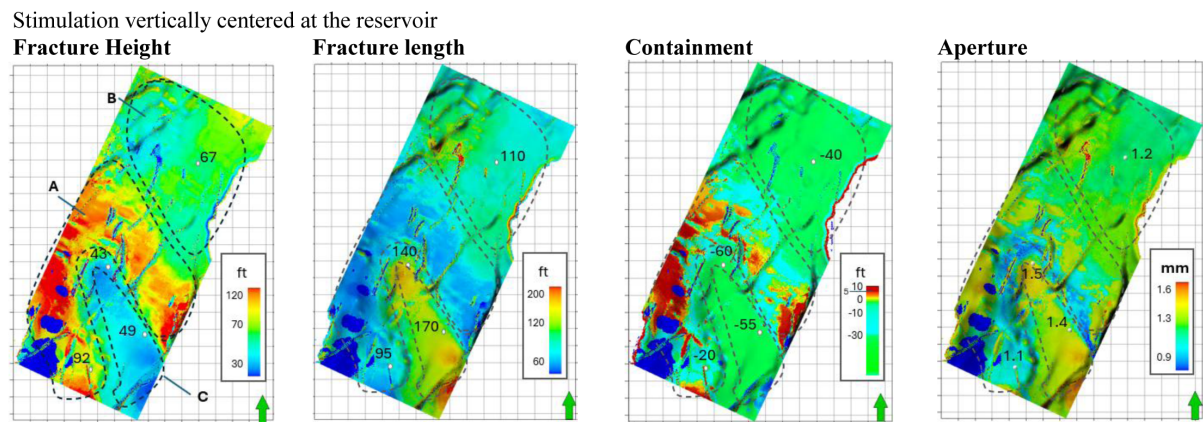


Figure 6—Height length, containment and fracture width for fracture initiation in a layer centered vertically in the reservoir (approximately), in-between stress barriers observed towards the south in the model.

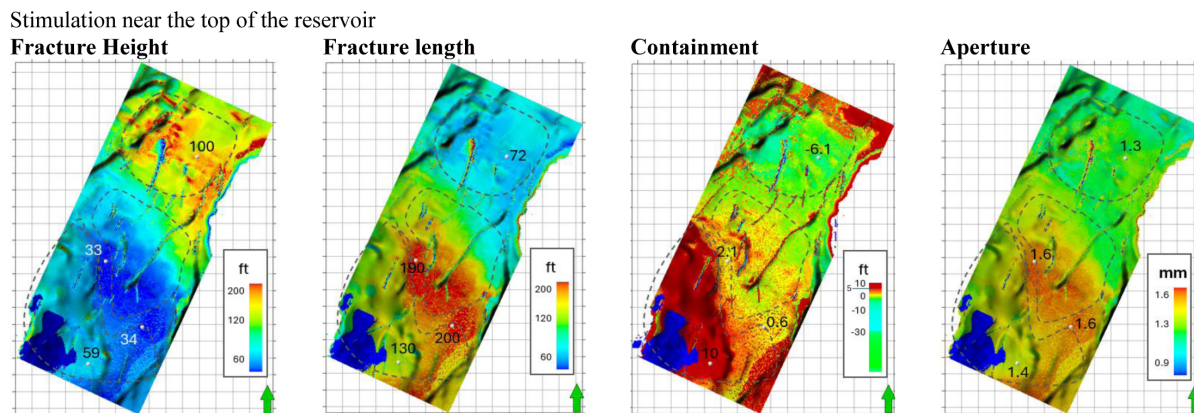


Figure 7—Height length, containment and fracture width for fracture initiation in a layer near the top of the reservoir. Fractures are shorter and longer than in Figure 6. Apertures also appear slightly higher and containment might be compromised in a large area in comparison with a stimulation scenario in deeper layers of the reservoir.

Figure 6 shows the results when the fractures are initiated near the middle of the reservoir, at a depth of approximately 80 ft below the horizon Pujama in Figure 4. Three areas with distinctive fracture height were observed. There is a strip (area A in the left frame) where the fracs are relatively taller than everywhere else, hence the length is relatively small (middle frame). Fracture growth in this area was predominantly upwards.

Containment in the third frame was computed as the depth difference between the top tip of the fracture and the upper boundary of the reservoir; i.e. positive values indicate that the fracture crossed to the upper reservoir. The results indicate that containment is good overall but there were patches in area A, where the model indicated that the top tip of the fracture crossed into the upper reservoir by more than 5 ft (1+ meter). The fractures that were contained broke through the upper internal barrier of the reservoir (above initiation point) but got arrested at the interface between the reservoir stimulated and the one above, where the stress is higher. If containment is deemed important, then area A appears risky. One may argue that less aggressive stimulation would contain the fractures but that would be at the expense of their size.

Towards the South (area C) the fractures were shorter, with average heights of the order of 40 ft to 50 ft. This height is smaller than the thickness of the reservoir (90-130 ft) and approximately corresponds to the distance between the stress barriers inside the reservoir (see Figure 4). In comparison to area A, the lateral fracture growth was larger in area C because the same volume was injected but as the vertical growth was constrained the fractures propagate deeper into the reservoir in the lateral direction. This might be considered positive depending on the case. For example, if there is a significant density of vertical natural fractures, it might be better to produce short but long fractures over taller and skinny ones. The area B (North) in the figure appears to have characteristics somewhere in the middle of those in area A and area C.

Figure 7 shows the results obtained when stimulation was modelled near the top boundary of the reservoir, approximately 22 ft below the seismic horizon Pujama. As in previous cases, three characteristic regions were observed. Overall, the fractures in this shallower configuration were both shorter and longer than those produced in the deeper stimulation scenario, suggesting greater variability in fracture geometry.

Containment was notably compromised in Area C (as shown in the figure), where simulations consistently indicated that the top tip of the fracture crossed into the upper formation by more than 10 ft (approximately 3 meters). In Area A, the containment results are more ambiguous. If interpreted in a binary sense—crossing versus non-crossing—containment would appear poor. However, the results suggest that the system is in a borderline regime. In this area, one cell might show clear upward fracture propagation by 5–10 ft, while an adjacent cell may indicate full containment. This sharp contrast between neighbouring cells gives the figure a pixelated appearance.

The presence of large apertures at the end of the job, combined with the prediction of long fractures, suggests some degree of ballooning. This, along with the limited upward growth, supports the idea that

vertical propagation is still strongly constrained under the given conditions. As in the previous case, the northern part of the field (Area B in the figure) stands out as different. The fact that fractures in this region are approximately 90 ft tall reflects the homogeneity of mechanical properties and in-situ stress.

To synthesize these results into a spatially coherent interpretation of completion favourability, the Completion Quality (CQ) Scheme described earlier can be applied.

Construction of the completion cubes. The very first step is deciding what variables we consider important and how important they are in relation to each other. This is mainly deciding the x_i variables and w_i weights in Equation 1. The choices made here are shown in the charts on the right of Figure 7.

The choices made in Figure 8 reflect a preference for large fractures, with heights around two-thirds of the reservoir thickness or more (Chart A), and for fractures that propagate deeply into the reservoir (Chart B), as indicated by a preferred height-to-length ratio of less than 1. Conversely, fractures with lengths comparable to the typical well spacing on a pad were penalized (Chart C). Relaxed stress regimes were favoured (Chart D). For all these variables, the weights w_i in Equation 1 were set to 1. Containment was not explicitly considered, as previous results indicated generally good containment across the model. The breakdown pressure—used as a proxy for the energy required to fracture the rock—was assigned a lower weight of 0.5, and high values were only lightly penalized (Chart E), given that the required pressures in this field are within attainable limits.

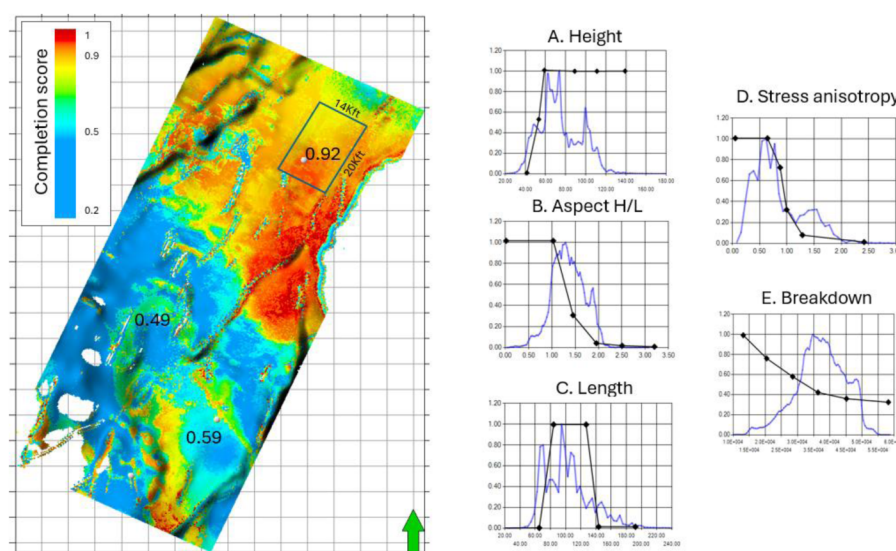


Figure 8—Completion quality cube for stimulation at a layer vertically centered in the reservoir. Five parameters were aggregated into a single score. The figure on the left shows the aggregated score. The charts on the right show the variables considered and the criteria imposed on them. The results shown were obtained while considering the values of the variables across a single layer studied.

No single variable dominated the final result. However, when a constraint was applied to limit fracture length, Area A in the figure was effectively discarded, with scores there dropping to approximately 0.45. If infinitely long fractures had been allowed, the score in that area would have increased slightly, to around 0.6. In either scenario, the northern part of the model consistently emerged as the most favourable area. The rectangle shown at the top of Figure 7 highlights a region that satisfies most of the imposed criteria (with a score near 0.9) and is large enough to accommodate multiple wells in a pad. This area could have been shifted slightly southward, but we deliberately avoided the faults in the model, whose effects were not explicitly accounted for.

Figure 9 shows that there is a noticeable difference between fracturing near the top of the reservoir and near its vertical center. However, the optimal areas for placing the first wells do not appear to strongly

depend on the depth of stimulation. Most differences are instead observed in the lower-scoring regions, where finer details play a greater role. To some extent, this outcome was expected, as the northern part of the model is relatively homogeneous compared to other areas.

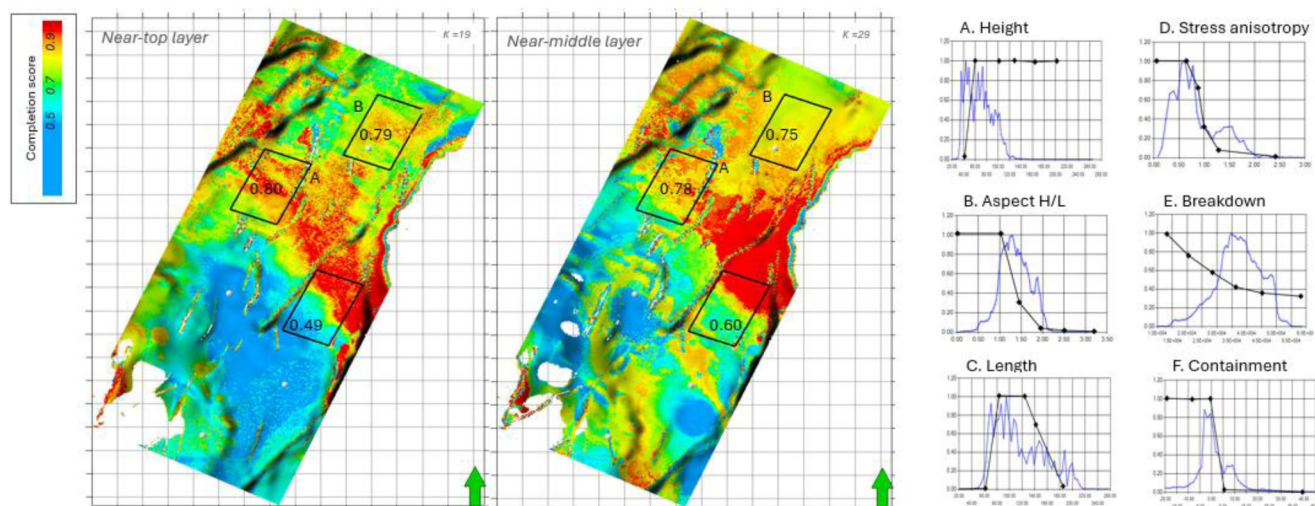


Figure 9—Comparison of two scenarios. Left: fractures were initiated near the top boundary of the reservoir. Right: Fracturing was carried-out across a layer near the vertical center of the reservoir. In both scenarios, the stimulation parameters were the same. A hydraulic fracturing simulation was done at every cell at the respective layer for the two investigated depths.

Overall, the results indicate that the internal structure of the reservoir plays a significant role in the potential success of hydraulic fracturing in this case. Beyond the specific findings for the example considered, this work also demonstrates the applicability of the method for large-scale reservoir screening. It enables the integration of dynamic fracture propagation properties, geomechanical parameters, and potentially an arbitrary number of other inputs into a unified evaluation framework.

Conclusions

This study introduced and demonstrated a method for large-scale screening of reservoirs to support well placement decisions. The approach was applied to a representative field example and successfully identified the most favourable areas for hydraulic fracturing, based on spatial variability in key reservoir properties.

The results also emphasized the role of internal reservoir structure as an influential variable, suggesting that features beyond traditional well-centric metrics can impact the effectiveness of stimulation. Over 150,000 simulations were executed as part of the analysis—a scale of computation that would be infeasible using localized, well-by-well modelling approaches.

The method provides a practical and physically consistent framework for field-wide evaluation, enabling early identification of promising regions and supporting more informed decisions during field development planning.

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